

$$1. P(\text{没有不合格品}) = \frac{C_{37}^2}{C_{40}^2} = \frac{\frac{1}{2} \times 37 \times 36}{\frac{1}{2} \times 40 \times 39} = \frac{111}{130}$$

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$$\Rightarrow P(\text{至少一件不合格品}) = 1 - \frac{111}{130} = \frac{19}{130}$$

主页: smallyangb88.github.io

$$2. P(A) = \left(\frac{1}{3}\right)^3 = \frac{1}{27} \quad P(B) = 1 \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{9} \quad P(C) = \frac{3!}{3^3} = \frac{2}{9}$$

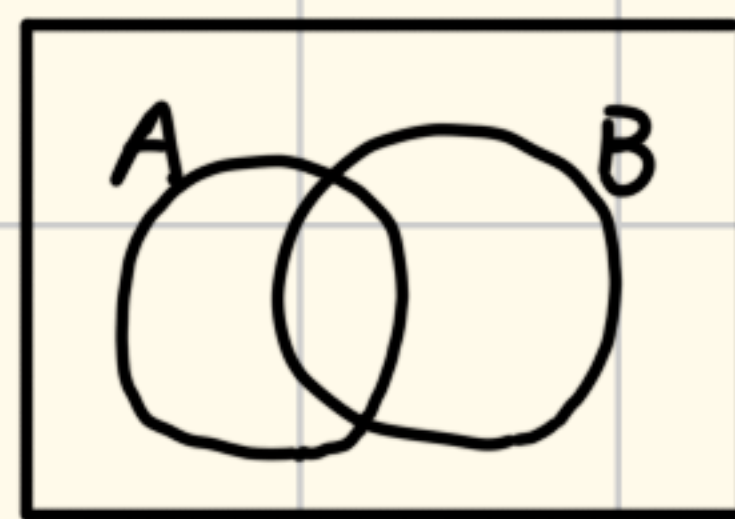
$$P(D) = 1 - P(B) = \frac{8}{9} \quad P(E) = \left(\frac{2}{3}\right)^3 = \frac{8}{27} \quad P(F) = \left(\frac{1}{3}\right)^3 = \frac{1}{27}$$

$$3. P(\text{两张均为黑桃}) = \frac{C_{13}^2}{C_{52}^2} = \frac{\frac{1}{2} \times 13 \times 12}{\frac{1}{2} \times 52 \times 51} = \frac{1}{17}$$

$$4. P(AB) = P(A) + P(B) - P(A \cup B) = p + q - r$$

$$P(A\bar{B}) = P(A) - P(AB) = r - q$$

$$P(\bar{A}\bar{B}) = P(\bar{B}) - P(A\bar{B}) = 1 - r$$



把这个图画出来
做这种题
就很清晰

5. A和B同时发生必然C发生, 翻译一下就是 $C \supset (A \cap B)$

$$\Rightarrow P(C) \geq P(A \cap B) = P(A) + P(B) - P(A \cup B) \geq P(A) + P(B) - 1$$

$$6. A_1 A_2 A_3 \subset A \Rightarrow P(A) \geq P(A_1 A_2 A_3) = P(A_1) + P(A_2) + P(A_3)$$

$$- P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3) + P(A_1 \cap A_2 \cap A_3)$$

$$\text{而 } P(A_1 \cap A_3) + P(A_2 \cap A_3) - P(A_1 \cap A_2 \cap A_3) = P((A_1 \cup A_2) \cap A_3)$$

$$\text{因此, 上式} = P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P((A_1 \cup A_2) \cap A_3)$$

$$\geq P(A_1) + P(A_2) + P(A_3) - 2$$

(同样地, 画出韦恩图就能很快知道该怎么做)

$$7. P(5人在不同层) = \frac{7 \times 6 \times 5 \times 4 \times 3}{7^5} = \frac{360}{2401}$$

$$8. P(\text{彼此不同}) = \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n} \quad (\text{前提是 } n \leq 365, \text{ 而 } n > 365 \text{ 时概率为 } 0)$$

$$P(\text{至少有两人相同}) = 1 - P(\text{彼此不同})$$

$$9. P(\text{中间号码为5}) = \frac{4 \times 1 \times 5}{C_{10}^3} = \frac{1}{6}$$

$$10. P(\text{每组都一黑一白}) = \frac{10 \times 10 \times 2 \times 9 \times 9 \times 2 \times \cdots \times 1 \times 1 \times 2}{20!} = \frac{2^{10} \cdot (10!)^2}{20!}$$

相当于是20个球都有编号,然后排成一列. 12, 34, 56这些相邻的算作一组.

最后每一组内还要乘2, 因为排队的时候同组黑白谁在前不确定

$$11. P(100条中有4条有记号) = \frac{C_{80}^4 C_{N-80}^{96}}{C_N^{100}}$$

对N的真值的最好的猜测应该是让这个概率取到最大值的N

$$\text{记 } a_N = \frac{C_{80}^4 C_{N-80}^{96}}{C_N^{100}} = C_{80}^4 \cdot \frac{(N-80)!}{96!(N-176)!} \cdot \frac{100!(N-100)!}{N!}$$

$$\text{则 } \frac{a_{N+1}}{a_N} = \frac{N-79}{N-175} \cdot \frac{N-99}{N+1} \quad \text{我们的目标是找 } a_N \text{ 的最大值}$$

求解 $\frac{a_{N+1}}{a_N} \geq 1$ 得 $N \leq 1999$. 所以N最好的估计是1999或2000

12. 基本事件空间 Ω 是所有可能出现的东西的集合

在这题里就是所有 C_{52}^6 种组合的集合

$$11) P(\text{含有黑桃K}) = \frac{C_{51}^5}{C_{52}^6}$$

$$12) P(\text{各种花色都有}) = P(1+1+2+2) + P(1+1+1+3) = \frac{C_4^2 (C_{13}^1)^2 (C_{13}^2)^2 + C_4^1 (C_{13}^1)^3 C_{13}^3}{C_{52}^6}$$

$$13) P(\text{至少有两张牌同点}) = 1 - P(\text{六张都不同点}) = 1 - \frac{C_{13}^6 \cdot 4^6}{C_{52}^6}$$

$$13. P(\text{一个小于} k \text{ 另一个大于} k) = \frac{C_{k-1}^1 C_{n-k}^1}{C_n^2} \quad \text{注意, 概率为0 不等价于不可能事件!}$$

14. (1)错 (2)错 (3)对 (4)错 (5)对. 因为 $P(A-B) = P(A) - P(AB)$ ($\forall A, B$)

$$15. \#(A) + \#(B) + \#(C) - [\#(AB) + \#(BC) + \#(AC)]$$

$$= \#(A \cup B \cup C) - \#(A \cap B \cap C) \leq \#(A \cup B \cup C) \leq \#(M)$$

$$16. \#(\text{巧} \cup \text{夹} \cup \text{冰}) = \#(\text{巧}) + \#(\text{夹}) + \#(\text{冰}) - \#(\text{巧} \cap \text{夹}) - \#(\text{巧} \cap \text{冰}) - \#(\text{夹} \cap \text{冰})$$

$$+ \#(\text{巧} \cap \text{夹} \cap \text{冰}) = 811 + 752 + 418 - 570 - 356 - 348 + 298 = 1005 \text{ 大于总人数}$$

17. 一方面, 不妨设 $P(A) \geq P(B)$. 则 $P(AB) - P(A)P(B) \leq P(B) - P(A)P(B)$

$$= P(B)(1 - P(A)) \leq \left(\frac{1 - P(A) + P(B)}{2} \right)^2 \leq \frac{1}{4}$$

另一方面, $P(A)P(B) - P(AB) = P(A)P(B) - P(A) - P(B) + P(A \cup B)$

$$= (1-P(A))(1-P(B)) - (1-P(A \cup B)) \leq \left(\frac{2-P(A)-P(B)}{2} \right)^2 - (1-P(A \cup B))$$

$$= \left(\frac{2-P(A \cup B)-P(A \cap B)}{2} \right)^2 - (1-P(A \cup B)) \leq \left(\frac{2-P(A \cup B)}{2} \right)^2 - (1-P(A \cup B))$$

$$= \left(\frac{1+t}{2} \right)^2 - t = \left(\frac{1-t}{2} \right)^2 \leq \frac{1}{4}$$

其中 $t = 1 - P(A \cup B) \in [0, 1]$

17题有个巧妙方法: 设 $X = \begin{cases} 1, & \text{若 } A \text{ 发生} \\ 0, & \text{若 } A \text{ 不发生} \end{cases}$, $Y = \begin{cases} 1, & \text{若 } B \text{ 发生} \\ 0, & \text{若 } B \text{ 不发生} \end{cases}$

则 $\begin{cases} E(X) = P(A) \\ D(X) = P(A)(1-P(A)) \end{cases}$. 且 $E(XY) = P(AB)$ ($XY = \begin{cases} 1, & \text{若 } A, B \text{ 同时发生} \\ 0, & \text{其它} \end{cases}$)

于是 $|P(A)P(B) - P(AB)| \leq \frac{1}{4}$ 于是 $|P(A)P(B) - P(AB)| = |E(X)E(Y) - E(XY)| \leq \frac{1}{4}$ (协方差的柯西不等式)

$$18. A = "1+1=3", B = "1+1=2" \quad \sqrt{D(X)D(Y)} = \sqrt{P(A)(1-P(A))P(B)(1-P(B))} \leq \frac{1}{4}$$

则 $P(A) = 0, P(B) = 1, P(AB) = 0$. 于是 A, B 不相容且独立

$$\begin{aligned} 19. P(A_{i_1} \cdots A_{i_p} \bar{A}_{j_1} \cdots \bar{A}_{j_q} U_{k_1} \cdots U_{k_r}) &= P((A_{i_1} \cdots A_{i_p}) \bar{A}_{j_1} \cdots \bar{A}_{j_q}) \\ &= P(A_{i_1} \cdots A_{i_p}) - \sum_{l=1}^q P(A_{i_1} \cdots A_{i_p} A_{j_l}) + \sum_{1 \leq l_1 < l_2 \leq q} P(A_{i_1} \cdots A_{i_p} A_{j_{l_1}} A_{j_{l_2}}) + \cdots + (-1)^q P(A_{i_1} \cdots A_{i_p} A_{j_1} \cdots A_{j_q}) \\ &= P(A_{i_1}) \cdots P(A_{i_p}) - \sum_{l=1}^q P(A_{i_1}) \cdots P(A_{i_p}) P(A_{j_l}) + \cdots + (-1)^q P(A_{i_1}) \cdots P(A_{i_p}) P(A_{j_1}) \cdots P(A_{j_q}) \\ &= P(A_{i_1}) \cdots P(A_{i_p}) (1 - P(A_{j_1})) \cdots (1 - P(A_{j_q})) = P(A_{i_1}) \cdots P(A_{i_p}) P(\bar{A}_{j_1}) \cdots P(\bar{A}_{j_q}) P(U_{k_1}) \cdots P(U_{k_r}) \end{aligned}$$

$$20. P(\text{最大等于 } m) = \frac{C_{m-1}^{k-1}}{C_n^k} \quad P(\text{最大不超过 } m) = \frac{C_m^k}{C_n^k}$$

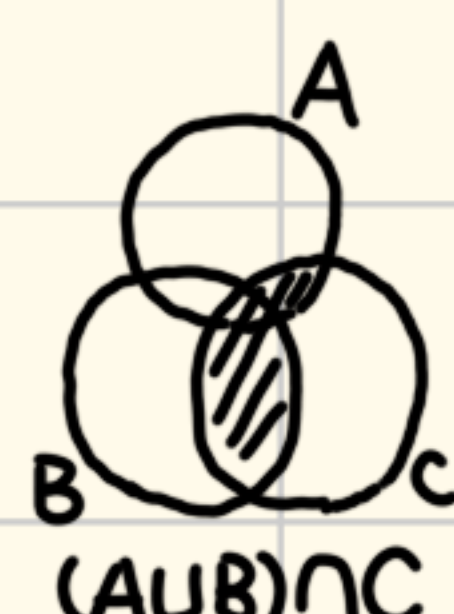
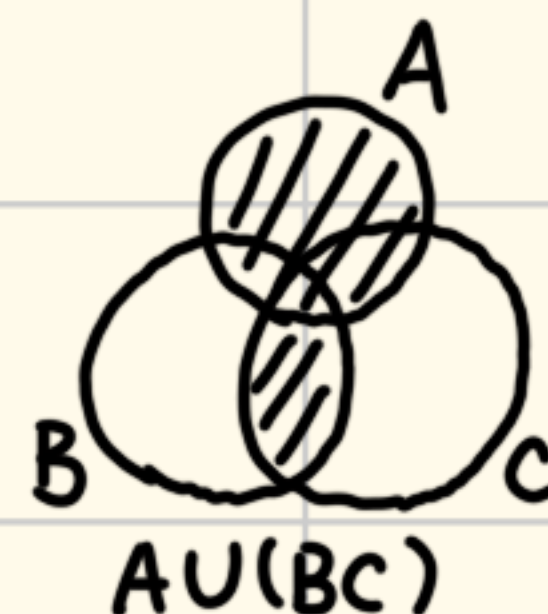
$$21. P(\text{选中4个}) = \frac{C_{45}^2 C_6^1}{C_{51}^3} \quad P(\text{选中5个}) = \frac{C_{45}^1 C_6^2}{C_{51}^3} \quad P(\text{选中6个}) = \frac{1}{C_{51}^3}$$

22. (1) 一定成立. 因为 $A \cup B = (A \setminus B) \cup B = (A \cap \bar{B}) \cup B$

(2) 不一定成立. 因为 $\forall a \in A$. 有 $a \notin \bar{A}B$ 但 $a \in A \cup B$

(3) 一定成立. 因为 $\overline{(A \cup B)} = \bar{A} \cap \bar{B}$

(4) 一定成立. 因为 $(AB) \cap (A\bar{B}) \subseteq B \cap \bar{B} = \emptyset$



(5) 不一定成立. 因为 $\forall a \in A \setminus (B \cup C)$. 有 $a \in A \cup (B \cap C)$ 但 $a \notin (A \cup B) \cap C$

$$23. \text{对 } n \in \mathbb{N} \text{ 归纳. } n=2 \text{ 时 } \#(A_1 \cup A_2) = \#(A_1) + \#(A_2) - \#(A_1 \cap A_2)$$

$$\text{假设 } n=m \text{ 时结论成立. 即 } \#(\bigcup_{i=1}^m A_i) = \sum_{k=1}^m (-1)^{k-1} \sum_{1 \leq i_1 < \cdots < i_k \leq m} \#(A_{i_1} \cdots A_{i_k})$$

$$\text{则 } n=m+1 \text{ 时 } \#(\bigcup_{i=1}^{m+1} A_i) = \#((\bigcup_{i=1}^m A_i) \cup A_{m+1}) = \#(\bigcup_{i=1}^m A_i) + \#(A_{m+1}) - \#((\bigcup_{i=1}^m A_i) \cap A_{m+1})$$

$$\text{而 } (\bigcup_{i=1}^m A_i) \cap A_{m+1} = \bigcup_{i=1}^m (A_i \cap A_{m+1}). \text{ 于是 } \# \left(\left(\bigcup_{i=1}^m A_i \right) \cap A_{m+1} \right) = \# \left(\bigcup_{i=1}^m (A_i \cap A_{m+1}) \right)$$

$$= \sum_{k=1}^m (-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq m} \# (A_{i_1} \cap A_{m+1} \cap \dots \cap A_{i_k} \cap A_{m+1}) = \sum_{k=1}^m (-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq m} \# (A_{i_1} \dots A_{i_k} A_{m+1})$$

$$\text{因此 } \# \left(\bigcup_{i=1}^m A_i \right) = \sum_{k=1}^m (-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq m} \# (A_{i_1} \dots A_{i_k}) \text{ 就是 } \sum_{k=1}^{m+1} (-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq m+1} \# (A_{i_1} \dots A_{i_k}) \text{ 中}$$

不包含 A_{m+1} 的项, 而 $\#(A_{m+1}) - \# \left(\left(\bigcup_{i=1}^m A_i \right) \cap A_{m+1} \right)$ 就是包含 A_{m+1} 的项

$$\text{所以 } \# \left(\bigcup_{i=1}^{m+1} A_i \right) = \# \left(\bigcup_{i=1}^m A_i \right) + \#(A_{m+1}) - \# \left(\left(\bigcup_{i=1}^m A_i \right) \cap A_{m+1} \right) = \sum_{k=1}^{m+1} (-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq m+1} \# (A_{i_1} \dots A_{i_k})$$

由归纳法原理知对所有 $n \in \mathbb{N}^* (n \geq 2)$ 命题成立

$$24. P(A|B) > P(A|\bar{B}) \Leftrightarrow \frac{P(AB)}{P(B)} > \frac{P(A\bar{B})}{P(\bar{B})} = \frac{P(A) - P(AB)}{1 - P(B)}$$

$$\Leftrightarrow P(AB)(1 - P(B)) > P(B)(P(A) - P(AB)) \Leftrightarrow P(AB) > P(A)P(B)$$

$$\Leftrightarrow P(AB)(1 - P(A)) > P(A)(P(B) - P(AB)) \Leftrightarrow \frac{P(AB)}{P(A)} > \frac{P(B) - P(AB)}{1 - P(A)} \Leftrightarrow P(B|A) > P(B|\bar{A})$$

$$25. P(\text{能得到}) = 1 - P(\text{在每个图书馆要么没有要么已借出}) = 1 - (1 - \frac{1}{4})^3 = \frac{37}{64}$$

$$26. \frac{P(\text{丙中} | \text{有两弹中})}{P(\text{丙不中} | \text{有两弹中})} = \frac{\frac{P(\text{丙中且两中})}{P(\text{两中})}}{\frac{P(\text{丙不中且两中})}{P(\text{两中})}} = \frac{0.4 \times [0.6 \times (1 - 0.5) + (1 - 0.6) \times 0.5]}{(1 - 0.4) \times 0.6 \times 0.5} = \frac{10}{9} > 1 \quad \text{丙中靶可能性更大}$$

$$27. P(\text{男} | \text{色盲}) = \frac{P(\text{男且色盲})}{P(\text{色盲})} = \frac{P(\text{男})P(\text{色盲} | \text{男})}{P(\text{男})P(\text{色盲} | \text{男}) + P(\text{女})P(\text{色盲} | \text{女})} = \frac{20}{21}$$

$$28. P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(A) + P(B) - P(A \cup B)}{P(B)} \geq \frac{P(A) + P(B) - 1}{P(B)} = 0.875$$

$$29. A \text{ 与 } B \text{ 独立且不相容} \Rightarrow P(AB) = P(A)P(B) \text{ 且 } P(AB) = 0 \Rightarrow \min\{P(A), P(B)\} = 0$$

$$30. P(C | \bar{A}\bar{B}) = \frac{P(\bar{A}\bar{B}C)}{P(\bar{A}\bar{B})} = \frac{P(\bar{A}C) - P(\bar{A}BC)}{P(\bar{A}) - P(\bar{A}B)} = \frac{(P(C) - P(AC)) - (P(BC) - P(ABC))}{(1 - P(A)) - (P(B) - P(AB))}$$

$$31. \text{考虑何时甲会赢. 假设甲赢时进行了 } k \text{ 局. 则 } (m \leq k \leq m+n-1)$$

则这 k 局里最后一局是甲赢的. 前 $k-1$ 局中甲任选 $m-1$ 局赢, 乙赢剩下 $k-m$ 局

$$\text{因此 } P(\text{甲赢}) = \sum_{k=m}^{m+n-1} C_{k-1}^{m-1} p^m (1-p)^{k-m} \stackrel{k=m+l}{=} \sum_{l=0}^{n-1} C_{l+m-1}^{m-1} (1-p)^l p^m$$

探索一下这个式子能否进一步化简.

$$\text{此处可借鉴公式 } \frac{1}{(1-x)^m} = \sum_{l=0}^{+\infty} C_{m+l-1}^{m-1} x^l \quad \left(\text{证法是对 } \frac{1}{1-x} = 1+x+x^2+\dots \text{ 两边求 } m-1 \text{ 次导} \right)$$

我们目标的形式和它很像,只是求和的指标不是到正无穷

我们仿照它的证法,自己操作一下.把 $+\infty$ 都换成 $n-1$ 即可

已有 $1+\lambda+\lambda^2+\dots+\lambda^{n-1}=\frac{1-\lambda^n}{1-\lambda}$. 两边求 $m-1$ 次导,再代入 $\lambda=1-p$,左边是待求式.

但是右边似乎没有继续操作的空间.所以我们猜它无法再化简了.

32. 不妨设甲先摸. 甲赢时,一定是甲乙轮流摸出若干个黑球,然后甲出一个白球

于是 $P(\text{甲赢}) = \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \frac{C_n^{2k}}{C_{m+n}^{2k}} \cdot \frac{n-2k}{m+n-2k}$ (这个式子是考虑选 $2k$ 个黑球放在最前面,

再选一个白球. 而课本答案的式子 $\frac{1}{C_{m+n}^n} \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} C_{m-1+n-2k}^{m-1}$ 则是考虑第一个白球在第 $2k+1$

位时,后 $m-1+n-2k$ 位选 $m-1$ 个白球. 两个式子等价,但似乎后者更好,因为在计

算的时候分母是统一的,只用算一次就行)

33. 一盒空而另一盒剩 r 根也就是共拿了 $2n-r$ 次,第一盒被拿了 n 次第二盒被

拿了 $n-r$ 次, 因此 $p = C_{2n-r}^n \left(\frac{1}{2}\right)^{2n-r}$

34. $P(\text{在第8只} | \text{不在前7只}) = \frac{P(\text{在第8只})}{P(\text{不在前7只})} = \frac{\frac{1}{8}p}{1-p+\frac{1}{8}p} = \frac{p}{8-7p}$

35. $P(\text{有一颗出现1点} | \text{三颗各不相同}) = \frac{P(\text{有一颗出现1点且三颗各不相同})}{P(\text{三颗各不相同})} = \frac{\frac{C_5^2 \cdot 3!}{6^3}}{\frac{C_6^3 \cdot 3!}{6^3}} = \frac{1}{2}$

36. $P(\text{最多只有一个坏了}) = C_3^1 \cdot 0.8 \cdot 0.2^2 + 0.2^3 = 0.104$

37. (1) $P(\text{失效}) = 1 - P(\text{每个都正常工作}) = 1 - \prod_{i=1}^6 (1-p_i)$

(2) $P(\text{失效}) = P(\text{三条路都失效}) = \prod_{i=1}^3 [1 - (1-p_{2i-1})(1-p_{2i})]$

38. $P(\text{下一代有}m\text{条}) = \sum_{k=m}^{+\infty} P(\text{有}k\text{个卵}) P(\text{孵化出}m\text{条} | \text{有}k\text{个卵})$

$= \sum_{k=m}^{+\infty} \frac{\lambda^k}{k!} e^{-\lambda} C_k^m p^m (1-p)^{k-m} = \left(\frac{p}{1-p}\right)^m e^{-\lambda} \sum_{k=m}^{+\infty} \frac{[\lambda(1-p)]^k}{k!} C_k^m$

$= \left(\frac{p}{1-p}\right)^m e^{-\lambda} \sum_{k=m}^{+\infty} \frac{[\lambda(1-p)]^k}{k!} \cdot \frac{k!}{m!(k-m)!} = \left(\frac{p}{1-p}\right)^m e^{-\lambda} \cdot \frac{1}{m!} \cdot [\lambda(1-p)]^m \sum_{k=m}^{+\infty} \frac{[\lambda(1-p)]^{k-m}}{(k-m)!}$

$= \left(\frac{p}{1-p}\right)^m e^{-\lambda} \cdot \frac{1}{m!} \cdot [\lambda(1-p)]^m \cdot e^{\lambda(1-p)} = \frac{1}{m!} (\lambda p)^m e^{-\lambda p}$

39. $P(\text{至少一男} | \text{至少一女}) = \frac{P(\text{至少一男且至少一女})}{P(\text{至少一女})} = \frac{\frac{2 \times 3}{8}}{1 - \frac{1}{8}} = \frac{6}{7}$

$$40. P(\text{已校正过}|\text{中靶}) = \frac{P(\text{已校正过且中靶})}{P(\text{中靶})} = \frac{\frac{5}{8} \times 0.8}{\frac{5}{8} \times 0.8 + \frac{3}{8} \times 0.3} = \frac{40}{49}$$

$$41. P(A) = P(B) = \frac{1}{2}, P(AB) = \frac{3}{8}. \text{由 } P(AB) \neq P(A)P(B) \text{ 知 } A, B, C \text{ 不相互独立}$$

$$42. P((A \cup B) \cap C) = P(A \cap C) + P(B \cap C) - P(A \cap B \cap C) \\ = P(A)P(C) + P(B)P(C) - P(A)P(B)P(C) = P(A \cup B)P(C)$$

$$P((AB) \cap C) = P(ABC) = P(A)P(B)P(C) = P(AB)P(C)$$

$$P((A-B) \cap C) = P((A-(A \cap B)) \cap C) = P(A \cap C) - P(A \cap B \cap C) \\ = P(A)P(C) - P(A \cap B)P(C) = P(A-(A \cap B))P(C) = P(A-B)P(C)$$

$$43. \text{投一次, } p = \frac{1}{6} \text{ 概率甲赢, } q = \frac{1}{8} \text{ 概率乙赢, } 1-p-q = \frac{7}{9} \text{ 概率无事发生}$$

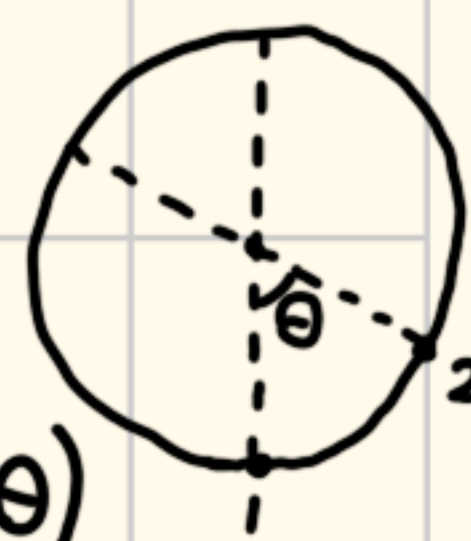
$$\text{于是 } P(\text{甲赢}) = \sum_{k=1}^{+\infty} p(1-p-q)^{k-1} = \frac{p}{1-(1-p-q)} = \frac{3}{4}$$

$$44. P(A \text{ 发生但 } B \text{ 不发生}) = \frac{1}{3}, P(AB \text{ 都不发生}) = \frac{1}{3}$$

$$P(A \text{ 在 } B \text{ 之前发生}) = \sum_{k=1}^{+\infty} (P(AB \text{ 都不发生}))^{k-1} \cdot P(A \text{ 发生但 } B \text{ 不发生}) = \frac{1}{2}$$

45. 思路如下. 第一个点随机取, 取好后固定. 第二个点也可以随机取, 这样前两个点可以构成一个圆心角 θ . 如果 $\theta \in (0, \pi)$, 那取第三个点的时候构成钝角三角形的概率就是 $\frac{2\pi - \theta}{2\pi}$. 所以我们用全概率公式来算

$$\text{即 } P(\text{构成钝角三角形}) = \int_0^\pi P(\theta \leq \text{圆心角} < \theta + \Delta\theta) P(\text{构成钝角三角形} | \theta \leq \text{圆心角} < \theta + \Delta\theta) \\ = \int_0^\pi \frac{2\Delta\theta}{2\pi} \cdot \frac{2\pi - \theta}{2\pi} d\theta = \int_0^\pi \frac{2\pi - \theta}{2\pi^2} d\theta = \frac{1}{2\pi^2} (2\pi\theta - \frac{1}{2}\theta^2) \Big|_0^\pi = \frac{3}{4}$$



$$46. P(A) = P(\text{至少6人遭遇不测}) = \sum_{k=6}^{2500} C_{2500}^k p^k$$

$$P(B) = P(\text{至多2人遭遇不测}) = \sum_{k=0}^2 C_{2500}^k p^k \quad (p = 0.0001)$$

$$47. P(\text{被击落} | \text{中两弹}) = \frac{P(\text{中两弹且被击落})}{P(\text{中两弹})} = \left(\frac{1}{10}\right)^2 + 2 \times \frac{1}{10} \times \frac{9}{10} + \left(\frac{2}{10}\right)^2 = \frac{23}{100}$$

$$48. (1) P(2 \text{ 人 } O) = C_5^2 \cdot 0.46^2 \cdot (1-0.46)^3$$

$$(2) P(3 \text{ 人 } O, 2 \text{ 人 } A) = C_5^2 \cdot 0.46^3 \cdot 0.40^2$$

(3) $P(\text{没有 } AB) = (1 - 0.03)^5$

49. 几何概型, 如右图所示. $p = \frac{1}{2} \times \frac{23}{24} \times \frac{23}{24} + \frac{1}{2} \times \frac{22}{24} \times \frac{22}{24} = \frac{1013}{1152}$

